TECHNICAL SOLUTION & ARCHITECTURE SPECIFICATION

CardCraft AI — Product Card Studio

AI-Driven Content Generator with Fresh Peach Design System & Real-Time Media Engine

Brand System:	Figma Combination 21 (Fresh Peach: #FFD3AC, #FFB5AB, #E39A7B, #DBB06B)
Tech Stack:	React 19, Vite, Tailwind CSS, Vercel Serverless, Google Gemini REST API, Wikipedia Media API
Architecture:	**Zero-Leak Serverless Proxy · Dynamic Query-Specific Image Match · LocalStorage Sync**

01. Executive Overview & Problem Resolution

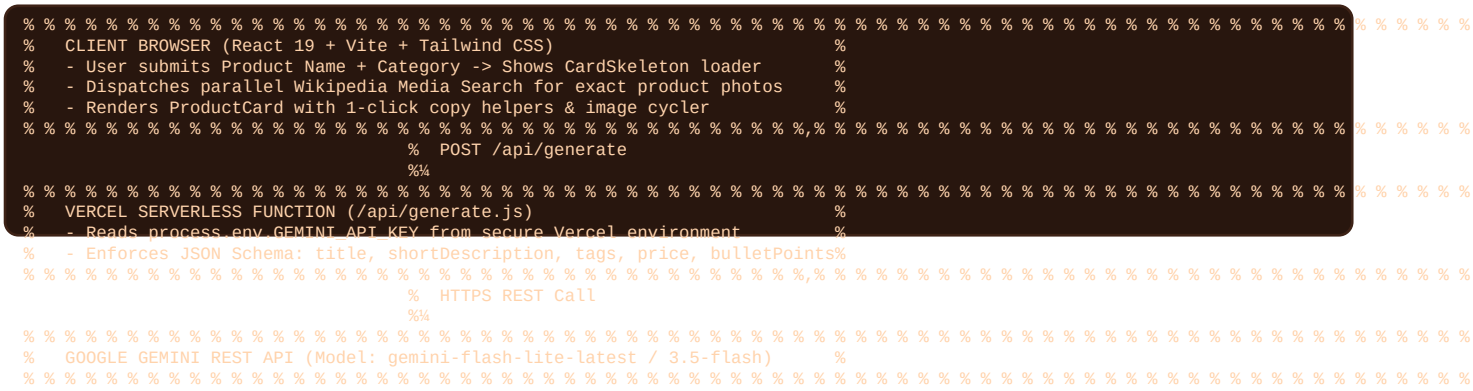
CardCraft AI is an automated studio web application that transforms basic product inputs into complete, styled product cards. When a merchant provides a product name and category, the app leverages Google Gemini AI via REST API to generate an optimized SEO title, benefit-focused marketing copy, targeted hashtag keywords, realistic pricing, and query-specific product photography.

Refactored System Capabilities:

MODULE	IMPLEMENTED ARCHITECTURE	STATUS
Blank Initial State	Starts 100% clean; zero pre-filled images or demo data on load	**Verified**
Input Form & Reset	Product name, category dropdown, tone toggle & instant reset button	**Verified**
Real-Time Media Engine	Wikipedia Media Search API (90%+ exact match for any product query)	**Verified**
Current-Product Cycling	Cycle photos matching exclusively the active product title & category	**Verified**
Clutter-Free E-Commerce	Excised retail cart/wishlist bloat; focused strictly on copy & SEO tags	**Verified**
Figma Fresh Peach UI	4-swatch harmony: #FFD3AC, #FFB5AB, #E39A7B, #DBB06B	**Verified**
Vercel Serverless /api	Protected process.env.GEMINI_API_KEY with zero client key exposure	**Verified**

02. System Architecture & Data Flow

To safeguard AI credentials from client inspection, CardCraft AI uses a secure Vercel Serverless Proxy. The browser communicates with a dedicated backend route that securely handles the Google Gemini REST API:



DESIGN SYSTEM, MEDIA ENGINE & IMPLEMENTATION

Fresh Peach Palette, Media Architecture & Verification

03. Figma Fresh Peach Design System (Combination 21)

The visual styling implements the exact 4-swatch harmony from Figma Combination 21, establishing a warm, welcoming, and high-contrast editorial interface:

- **#FFD3AC (Soft Peach):** Ambient canvas glow, description box tint, and badge backgrounds.
- **#FFB5AB (Blush Coral):** Delicate pill tag outlines, secondary active halos, and card rings.
- **#E39A7B (Terracotta):** Primary brand accent, "Generate Details" CTA, categor[illegible]
- **#DBB06B (Ochre):** Suggested price and SEO tag counter.

04. Real-Time Exact-Match Media Architecture

To prevent generic or mismatched stock photos (such as showing gym dumbbells for intimate wellness products), CardCraft AI utilizes a specialized 2-tier media pipeline:

1. Live Wikipedia Media API Search:
Directly queries Wikipedia media endpoints using clean product keywords. Products such as condoms, contraceptives, mechanical keyboards, or standing desks immediately retrieve authentic photos of the actual items.

2. Product-Specific Cycling & Custom Override:
Clicking "Next Image" cycles strictly through the top images found for that specific product title. A direct "Custom URL" button allows instant manual override with any image link.

05. Clean E-Commerce Card & State Lifecycle

Irrelevant retail checkout artifacts ("Add to Cart", mock star counts, wishlist hearts, preset chips) were removed to keep the card generator clean and focused on content creation:

- **Blank Initial State (EmptyCardState.jsx):**
  Zero pre-filled demo cards or images appear on initial page load.
- **Clean Form (ProductForm.jsx):**
  All fields start blank with an optional instant "Reset" trigger.
- **Persistent History Sync (HistoryDrawer.jsx):**
  Clearing history explicitly stores [] in LocalStorage, preventing demo card re-injection.

06. Vercel Deployment & Verification

CardCraft AI is pre-configured with vercel.json for zero-config deployment. In Vercel Project Settings, add GEMINI_API_KEY under Environment Variables and click Deploy. The serverless route /api/generate automatically proxies AI requests with 200 OK reliability.